

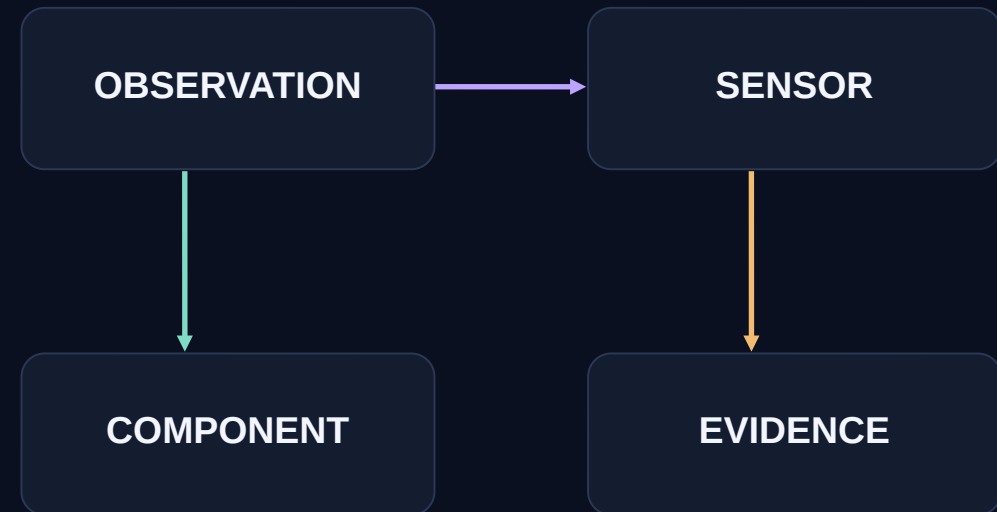
From signals to evidence.

RDF + telemetry context + SPARQL, learned through a working application.

A REAL LOCAL APPLICATION

Signal Atlas

One investigation. Seven sources.
Twelve executable questions.



Which results need an evidence review?

A signal can look normal while the evidence supporting its interpretation is incomplete.

| THE INVESTIGATION

Find observations whose sensor calibration did not cover the full sampled window.

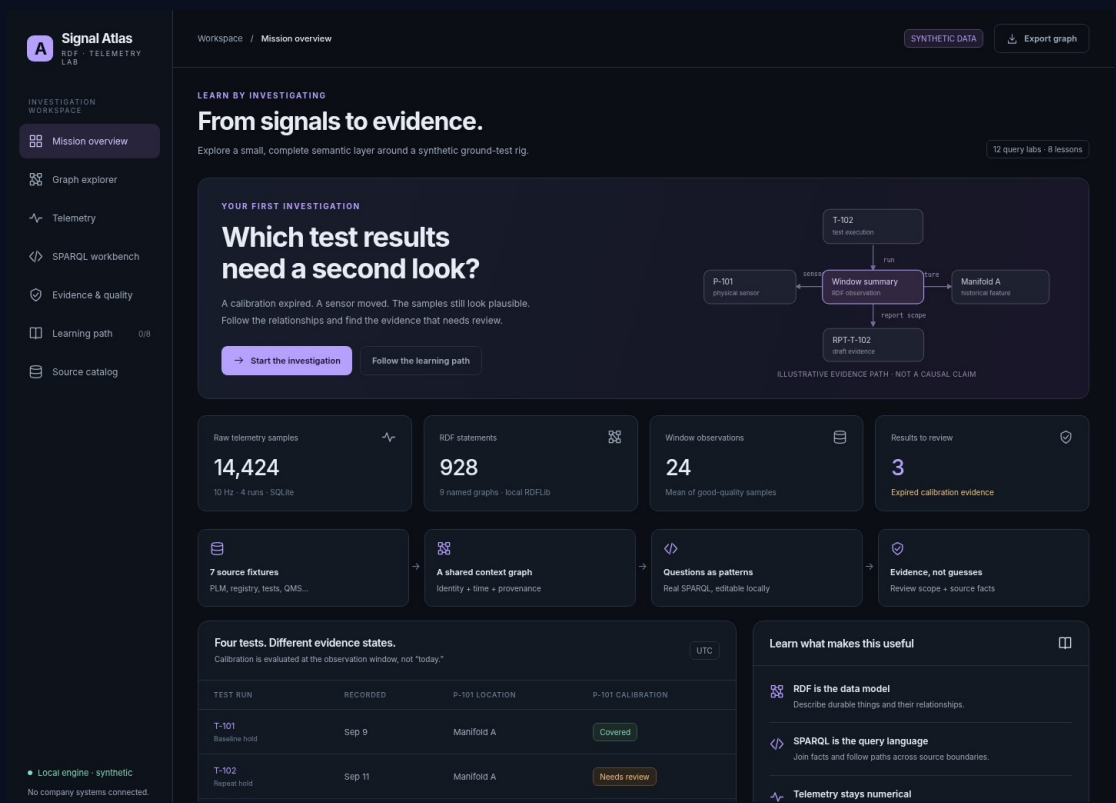
| THE PAYOFF

Explain the sensor, its location at that time, the affected report and the exact source evidence.

Not: “Is this hardware safe to fly?”

A workbench, not just a graph picture.

Seven pages connect exploration, query editing, raw values, evidence and learning.



EXPLORE

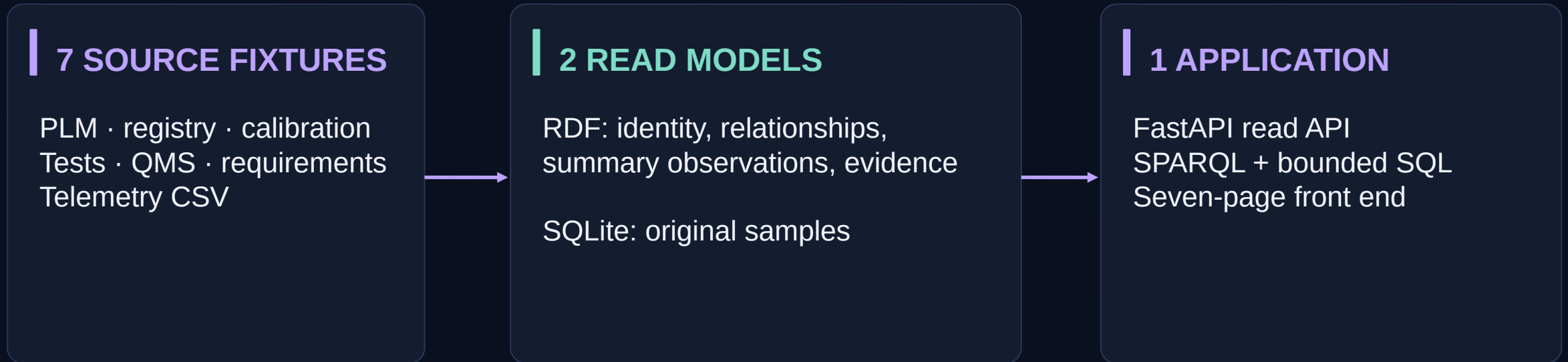
Follow actual RDF edges and inspect their source graph.

EXECUTE

Edit SPARQL and run it against the local RDFLib dataset.

Keep numeric scale and semantic context separate.

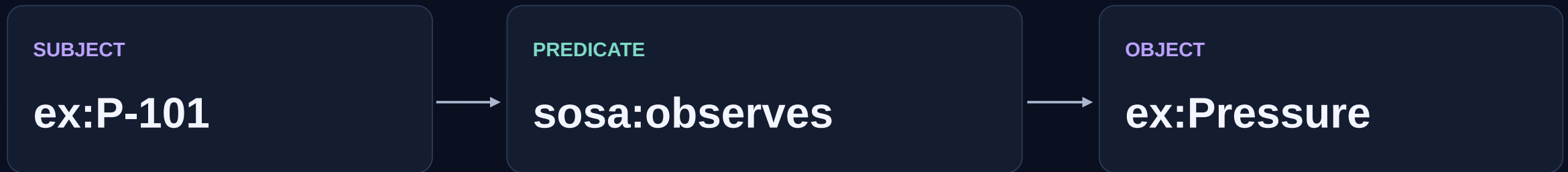
This is a design choice for the lab, not a limitation of RDF.



At work: retain the existing governed warehouse / telemetry platform.

A triple is a typed statement.

Subject → predicate → object. Resource identifiers are not just display labels.



```
ex:P-101 rdfs:label "Primary pressure sensor" .
```

An IRI names a resource. A literal carries a value, such as text, a number or a date.

14,424 samples. Just 24 observation summaries.

The graph points back to the precise sample window instead of hiding the raw data.

14,424

RAW SAMPLE ROWS

24

RDF WINDOW SUMMARIES

928

BASELINE STATEMENTS

observation → run + channel + time window
→ sensor + component + procedure + unit
→ source evidence + raw data selector

Use familiar terms before inventing an ontology.

A small custom namespace handles the local relationships the exercise needs.

| SOSA

Sensor
Observation
Observed property
Feature of interest

| PROV + UNITS

Source derivation
Generation time

Explicit QUDT unit IRIs

| ex: LAB TERMS

TestRun · Channel
Deployment · Calibration
Requirement · Report
partOf · validity bounds

References: [w3.org/TR/vocab-ssn/](https://www.w3.org/TR/vocab-ssn/) · [w3.org/TR/prov-o/](https://www.w3.org/TR/prov-o/) · qudt.org/vocab/unit/

Sensor $\neq$ channel $\neq$ deployment.

The distinction prevents a current location from rewriting historical observations.

PHYSICAL SENSOR

P-101

The same instrument persists across the move.

ACQUISITION CHANNEL

CH-P-101

A stream definition with a raw tag and declared unit.

DEPLOYMENT

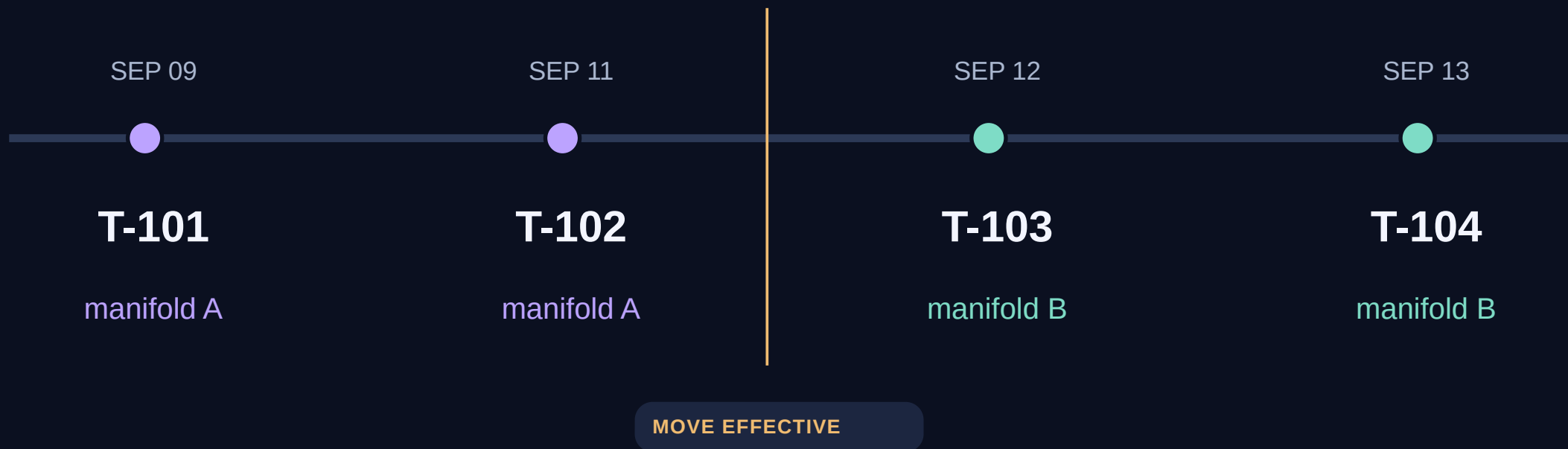
DEP-P-101-A / B

Where the instrument was installed, with effectivity.

Source IDs need explicit mappings and owners. Similar names are not proof of identity.

The same sensor. A different component.

P-101 moves at 2026-09-12 00:00 UTC. Each observation preserves its measurement-time context.



Do not join every past observation to “where the sensor is today.”

“Which source said this?” is a queryable question.

Source grouping, file hashes and mapping versions travel with the graph.

```
GRAPH ?sourceGraph {  
  ex:P-101 ?predicate ?object .  
}
```

| SOURCE CATALOG

- Source file + SHA-256
- Snapshot time + mapping version
- Synthetic / live status
- Authority remains a separate question

A hash identifies bytes. It does not prove truth, completeness or permission.

SPARQL joins by sharing variables.

Read each pattern as a statement with a blank to fill.

```
PREFIX sosa:<http://www.w3.org/ns/sosa/>
```

```
SELECT ?sensor ?property WHERE {  
  ?sensor a sosa:Sensor .  
  ?sensor sosa:observes ?property .  
}  
ORDER BY ?sensor
```

?sensor

The same variable connects the two patterns.

EXPECTED

Six sensor rows in the supplied baseline.

A path finds reachable things; it does not infer a cause.

The + operator follows one or more edges. The * operator also permits zero hops.

```
SELECT ?component WHERE {  
  ?component :part0+ ex:rig-01 .  
}
```

TRY IT

Run query 03: 4 descendants.
Change + to *: 5 results, including rig-01 itself.

Traversal $\neq$ automatic inference $\neq$ physical causality

Coverage must include the whole sampled window.

Certificate validity is [validFrom, validTo). The final sample is still a measured instant.

validFrom $\leq$ first sample AND validTo $>$ last sample

THE BASELINE

P-101 expires September 10.
T-101 is covered.
T-102 / T-103 / T-104 need review.

THE COUNTEREXAMPLE

An approved replacement covers the window.
The old expired record must not keep raising an alert.

Find expired evidence, then exclude covering replacements.

Query 05 expresses the domain rule. The full executable query is in the app and training guide.

```
?cert ex:sensor ?sensor ;  
    ex:validTo ?expiry ;ex:status "approved" .  
FILTER (?expiry <= ?end)  
FILTER NOT EXISTS {  
    ?cover ex:sensor ?sensor ;ex:status "approved" ;  
    ex:validFrom ?from ;ex:validTo ?to .  
    FILTER (?from <= ?start && ?to > ?end)  
}
```

3

review candidates

Expired, unknown and structurally incomplete are different.

Do not collapse uncertainty into a false “pass / fail” label.

3 · EXPIRED EVIDENCE

P-101 observations have known expired records and no covering approved replacement.

4 · UNKNOWN EVIDENCE

P-103 observations have no certificate in this snapshot. That is not proof calibration never happened.

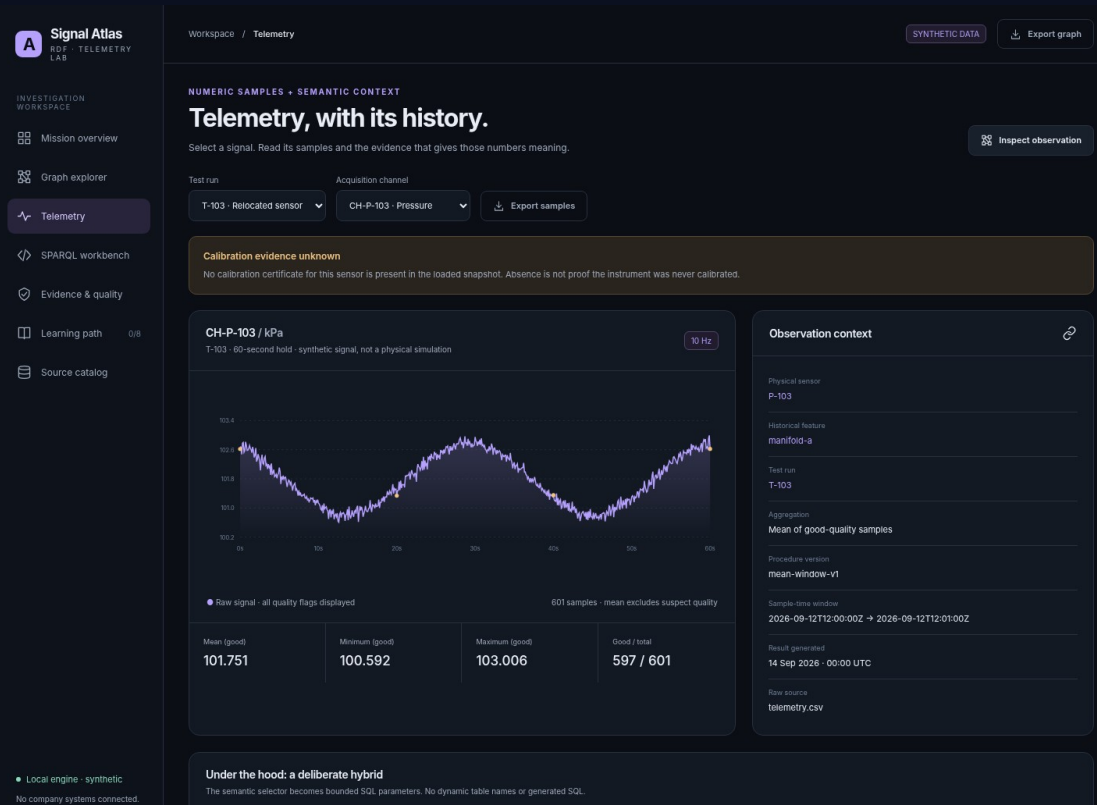
1 · UNIT CONTRACT

CH-AUX-401 lacks a declared unit in the channel registry. Raw voltage samples still exist.

These counts have different grains: observations, observations, channel.

Move from a relationship to the original samples.

Select T-103 / CH-P-103 to see why “unknown evidence” does not erase the signal.



601 SAMPLES

The detailed values remain in the numeric store.

597 GOOD

Four suspect samples are retained; the mean excludes them.

SELECT returns rows. ASK returns a Boolean. CONSTRUCT returns a graph.

A returned graph is not a write-back operation.

| SELECT

Which observations?

Result bindings you can inspect or tabulate.

| ASK

Does a condition match?

True / false against the selected dataset.

| CONSTRUCT

What review graph follows?

15 temporary triples for 3 review candidates.

SHACL asks whether the graph satisfies declared shapes.

The lab evaluates an explicitly limited subset; it is not a full SHACL implementation.

```
ex:ChannelShape a sh:NodeShape ;  
  sh:targetClass ex:Channel;  
  sh:property [  
    sh:path ex:unit;  
    sh:minCount 1; sh:maxCount 1;  
    sh:nodeKind sh:IRI  
  ].
```

BASELINE

One violation: a missing unit on CH-AUX-401.

LIMIT

Supported shapes only. Unknown constructs fail closed. No truth or safety guarantee.

Change a copy. Preserve the evidence.

Both scenarios recompute an isolated graph; neither overwrites the baseline.

| COVERING CALIBRATION

$3 \rightarrow 0$

Expired-evidence matches disappear in the hypothesis only.
P-103 remains unknown.

| DECLARE THE UNIT

$1 \rightarrow 0$

The structural violation disappears in the hypothesis only.
No operational acceptance follows.

Three edits make the mental model stick.

Use the real app for these edits; the offline HTML only replays saved results.

01 · PATH DEPTH

Query 03: change + to *.

Predict the extra result before executing.

02 · MISSING DATA

Query 04: make the OPTIONAL unit relationship required.

Which observations vanish?

03 · SOURCE CONTEXT

Query 08: inspect the named graph and compare it with the source catalog.

What does the hash prove?

Use a graph when relationships earn their complexity.

SQL can answer these questions too. The choice is not a contest between query languages.

| STAY WITH SQL

- Large numeric scans
- Stable tabular joins
- Aggregates and time-series analysis
- Existing governed marts

| CONSIDER RDF

- Shared cross-system identities
- Reusable relationship meaning
- Changing traversal questions
- Explainable evidence links

Expand by useful question, not by system count.

These are hypotheses to validate with the SME, not claims about existing company capability.

CALIBRATION IMPACT

Which downstream evidence needs review?

CHANNEL DISCOVERY

Find signals by property, unit and historical component.

COMPARISON ELIGIBILITY

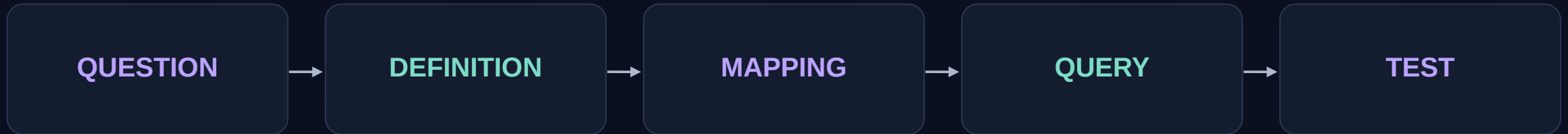
Select windows with approved contextual compatibility.

REPORT LINEAGE

Trace a report or requirement to exact source observations.

Turn the expert's explanation into executable knowledge.

The target is a second person who can reproduce the answer and explain its limits.



For every rule, capture a counterexample and an owner.

Transfer the synthetic prototype one way.

Preserve existing Paul OS, governed data access and company security patterns.

DISCOVER

Read the real work repository.
Resolve source owners, IDs,
event time, effectivity and
coverage.
Do not invent connections.

GATE

Authentication and authorization
Graph / sample / inference
access
Query limits and audit
Approved source slice

PILOT

Curated read-only questions
Evidence and freshness labels
SME-reviewed counterexamples
Second-person teach-back

What has been checked—and what has not.

Use the validation report rather than inferring production readiness from a polished interface.

| EXECUTED

- 43 automated tests passed
- 9 browser interaction checks
- All seven pages at phone width
- Actual RDFLib queries and SQL samples

| NOT CLAIMED

- Remote GitHub repository or deployment
- Live company adapters or authentication
- Docker / GitHub Actions execution
- Full SHACL conformance or pySHACL run

Start with the app. Finish with a teach-back.

The best next step is a narrow investigation with the single expert—not an enterprise ontology project.

FIRST SESSION

Open the lab → run query 05
Follow the historical deployment
Inspect original samples and sources
Change one query and explain the result

PRIMARY REFERENCES

RDF: [w3.org/TR/rdf11-concepts/](https://www.w3.org/TR/rdf11-concepts/)

SPARQL: [w3.org/TR/sparql11-query/](https://www.w3.org/TR/sparql11-query/)

SOSA: [w3.org/TR/vocab-ssn/](https://www.w3.org/TR/vocab-ssn/)

PROV: [w3.org/TR/prov-o/](https://www.w3.org/TR/prov-o/)

SHACL: [w3.org/TR/shacl/](https://www.w3.org/TR/shacl/)